

PAPER_1126_UPDATE_PSR_J0030_NICER_Astrophysics

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PAPER_1126_UPDATE — PSR J0030 NICER + Astrophysics Extensions

Original: PAPER_1126 Astrophysics Catalog

Update: Session June 2026

Status: Update

Session Extensions

NICER PSR J0030+0451 anchor values ($M = 1.4 M_{\text{sun}}$, $R = 13 \text{ km}$) wired:

- Compactness $GM/(Rc^2) = 0.1593$
- Schwarzschild factor 0.3185
- UQFF R correction: $1 + \text{BETA}_I \cdot \text{TRZ} \cdot \text{PHI} \cdot \text{SSQ}$ giving $R_{\text{uqff}} = 13.375 \text{ km}$ (2.9% from NICER)
- TOV limit $2.5 M_{\text{sun}}$
- GW170817 $M_{\text{max}} 2.17 M_{\text{sun}}$

Plus 19 additional astrophysical systems (Crab, Eta Car, Westerlund2, Lagoon, Orion, NGC3603, Rings) wired in `calculate_astrophysics`.